

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Data Sheet for Induction Bends
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006
Document Revision	R01
Document Status	Issued For Review
Originator	E. Uwaoma
Security Classification	Restricted
Issue Date	11 July 2022

Data Sheet for Induction Bends

R01	11.07.22	Issued For Review	E. Uwaoma	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	11/07/2022	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

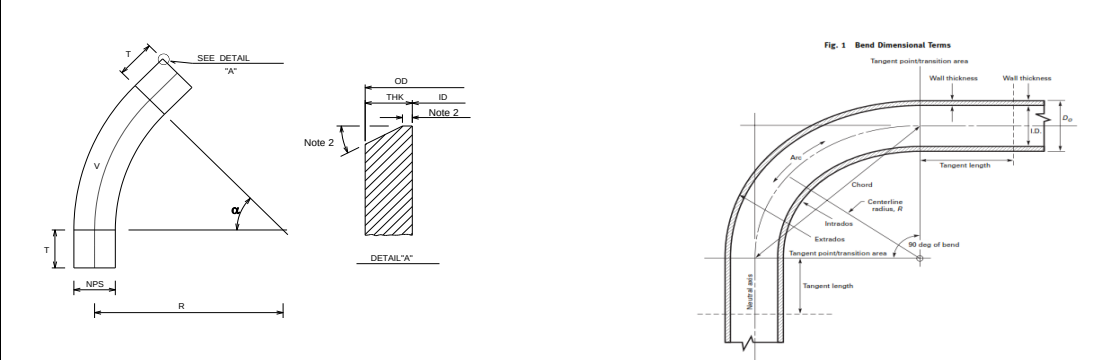
The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

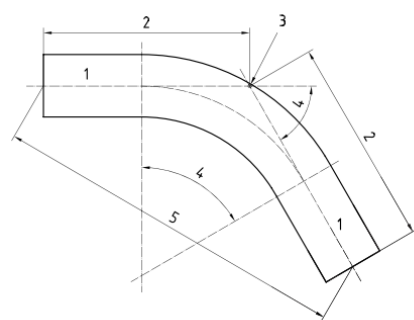
KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

Data Sheet for Induction Bends

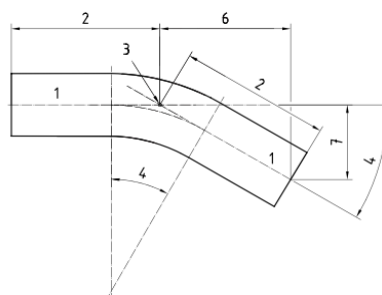
Document No.	Description

1	Project Name	Gas Gathering Pipelines & Associated Infrastructure- Phase 1		
2	Document Title	Data Sheet for Induction Bends	Revision Code.	R01
3	Document No.	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006	Revision Status	Issued for Review
4	Bend Characteristics			
5	Fluid	Multiphase (Gas and Condensate)		
6	Mother Pipe Material	Refer to Table 2 and 3		
7	Mother Pipe NPS	Refer to Table 2 and 3		
8	Manufacturing Process	Hot Induction Bending		
9	Minimum Bend Radius	5*D, Tolerance of ±10 mm		
10	Pressure Rating	900#		
11	Bend Angle	Refer to Table 2 and 3. Tolerance of ±1°		
12	Min.Tangent Lengths (mm)	500		
13	Chemical composition	Shall comply with mother pipe requirement		
14	Mother Pipe Diameter Internal Diameter (mm)	TBA		
15	Mother Pipe Diameter Outside Diameter (mm)	Refer to Table 2 and 3		
16	Mother Pipe Wall Thickness (mm)	Refer to Table 2 and 3		
17	Applicable Design Code	DEP 31.40.20.33-Gen / DEP 31.40.20.37-Gen and ISO 15590-1.		
18	Design Conditions			
19	Design Pressure (Barg)	110.00		
20	Design temperature (°C)	60.00		
21	Corrosion Allowance (mm)	Refer to tables 2 and 3		
22	Design Factor	Refer to tables 2 and 3		
23	Hydrotest Presure (Barg)	137.5		
24	Other Requirements			
25	Bend End Finish	Bevel Ends		
26	External Coating	3LPE As per DEP 31.40.30.31-Gen		
27	Bevel Protector	PVC Vinyl		
28	Matching Pipe (straight)	Pipes as per Datasheet for Line Pipe		
29	Dimensions			
30	Out-of-roundness	1% for bend end and 2.5% for bend body		
31	Out-of-planeness	1 % Of OD As per DEP 31.40.20.33-Gen		
32	End out-of-squareness	Tolerance of ±3mm		
33	Bevel End	30° with tolerance as provided for the pipe		
34	Testing & Inspection / Acceptance Requirements			
35	Qualification	As per DEP 31.40.20.37-Gen		
36	Ultrasonic	Ultrasonic test shall be performed on the outer surface of bends if required		
37	Magnetic Particle Inspection	As per DEP 31.40.20.37-Gen		
38	Hardness (maximum)	As per ISO 15590-1 Sec. 9.5.4, Specification for Induction bends		
39	Residual magnetism	As per DEP 31.40.20.37-Gen		
40	The line pipe for the Induction Bend shall be manufactured and tested in accordance with Specification for Line Pipe: EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001 and Specification for Induction Bend: EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006			
41	Marking			
42	Marking (location)	Both ends of each bend As per DEP 31.40.20.37-Gen		
43	Marking	As per DEP 31.40.20.37-Gen		
44	Standard Marking	As per DEP 31.40.20.37-Gen		
45	Bend Length Calculation			
46	Bend Length, L_b (Formula)		(((Bend Radius x Bend Angle x pi/180)) + (2xtangent length))/1000)	
47	Bend Angles (variable)	Degree	Refer to tables 2 and 3	
48	Bend Length (including tangent length)	m	To be calculated by Vendor.	
49	Quantities (See Note 3)	Pcs	Refer to tables 2 and 3	
50	Required No. of Bends - refer to Pipeline Material Take-Off (MTO) Report:			
51	Note			
52	1. Coating of the Induction Bends shall be in accordance with Technical Specification for Line Pipe Coating			
53	2. Minimum design temperature shall be 0 deg C, unless stated otherwise on the purchase order			
54	3. Quantities shall be confirmed after Detailed Topographical Survey			
55	Fig 1: Typical Bend Detail			
56	 Fig 1 Bend Dimensional Terms			
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a) Bends with angles of 15° and greater



b) Bends with angles of less than 15°

Key

- 1 Centreline axis
- 2 Centre to end
- 3 Centre of bend
- 4 Bend angle
- 5 Chord
- 6 Extension
- 7 Offset

Table 1: REFERENCES:

REF.	DOCUMENT NUMBER	TITLE
1	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-DWG-001	Overall Location Plan Map / Satellite Imagery
3	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Pipeline Wall Thickness Calculations
4	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-001	Data Sheet for Line Pipe
5	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006	Specification for Induction Bends
6	DEP 31.40.30.31-Gen February 2017	Pipeline External Coatings (Amendments/Supplements to ISO 21809-1:2011 and ISO 21809-2:2014)
7	DEP 31.40.20.33-Gen. February 2017	Line pipe induction bends (amendments/supplements to ISO 15590-1) Line pipe for critical service (amendments/supplements to IOGP S-616 and DNVGL-ST-F101 2017)
8	DEP 31.40.20.37-Gen February 2019	Petroleum and natural gas industries — Induction bends, fittings and flanges for pipeline transportation systems - Part 1: Induction Bend
9	ISO 15590-1	Gas Transmission and Distribution Piping Systems
10	ASME B31.8	

Table 1: 24" INDUCTION BEND FOR PIPELINE FROM BUBOUWE MANIFOLD STATION TO ELEPA MANIFOLD STATION

SN	ANGLE (DEGREE)	KP (Km)	Mother Pipe Material	Wall Thickness (m)	Outside Diameter (mm)	Design Factor	Corrosion Allowance (mm)	NORTHING (m)	EASTING (m)	LOCATION
1	90 (2)	0+000	API 5L X70 PSL 2	15.88	610	0.6	3.00	494208.76	179066	BUBOUWE MANIFOLD STATION
2	75	0+251	API 5L X70 PSL 2	15.88	610	0.6	3.00	494371.82	179257	INFIELD
3	90	6+988	API 5L X70 PSL 2	15.88	610	0.6	3.00	490530.89	184791	INFIELD
4	35	7+067	API 5L X70 PSL 2	15.88	610	0.6	3.00	490596.55	184837	INFIELD
5	90	7+314	API 5L X70 PSL 2	15.88	610	0.6	3.00	490842.14	184836	INFIELD
6	90 (2)	7+321	API 5L X70 PSL 2	15.88	610	0.6	3.00	490843.04	184819	ELEPA MANIFOLD STATION

Table 2: 36" INDUCTION BEND FOR PIPELINE FROM BUBOUWE MANIFOLD STATION TO ELEPA MANIFOLD STATION

SN	ANGLE (DEGREE)	KP (Km)	Mother Pipe Material	Wall Thickness (m)	Outside Diameter (mm)	Design Factor	Corrosion Allowance (mm)	NORTHING (m)	EASTING (m)	LOCATION
1	90 (2)	0+000	API 5L X70 PSL 2	23.83	914	0.6	6.00	490847.44	184829	ELEPA MANIFOLD STATION
2	90	0+010	API 5L X70 PSL 2	23.83	914	0.6	6.00	490847.39	184840	INFIELD
3	55	0+361	API 5L X70 PSL 2	23.83	914	0.6	6.00	490496.97	184840	INFIELD
4	34	3+581	API 5L X70 PSL 2	23.83	914	0.6	6.00	488660.92	187485	INFIELD
5	51	14+855	API 5L X70 PSL 2	23.83	914	0.6	6.00	488421.13	198757	INFIELD
6	34	19+356	API 5L X70 PSL 2	23.83	914	0.6	6.00	484882.92	201539	INFIELD
7	32	25+227	API 5L X70 PSL 2	23.83	914	0.6	6.00	483106.54	207135	INFIELD
8	30 (2)	31+376	API 5L X70 PSL 2	23.83	914	0.6	6.00	478451.28	211159	AKABEL MANIFOLD STATION